OAVS • TGT RECRUITMENT EXAMINATION

Computer Science Teacher

MCQ Question Bank — Complete Practice Set

Topic: Modeling Techniques

126	11	100%
TOTAL MCQS	SUB-TOPICS	ANSWERS + EXPLANATIONS

Strictly aligned to the OAVS TGT Computer Science — Modeling Techniques syllabus.

Installment 1 of the question bank — additional sets available on request.

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Chapter 1: Fundamentals of Data Models

15 Questions

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Q1 | Definition-based | Definition of Model

A data model in a database system is best defined as:

(A) A programming language used to write queries

(B) A conceptual tool that describes data, data relationships, and constraints

(C) A hardware component of the DBMS

(D) A physical storage device for data

Correct Answer: B

Explanation: A data model is a conceptual framework that defines how data is structured, related, and constrained within a database, independent of physical storage.

Q2 | Concept-based | Purpose of Data Modeling

The primary purpose of data modeling is to:

(A) Encrypt sensitive data automatically

(B) Replace the need for a DBMS

(C) Increase the physical size of the database

(D) Provide a blueprint for organizing and understanding data before implementation

Correct Answer: D

Explanation: Data modeling creates a structured blueprint that helps designers understand data requirements and relationships before building the actual database.

Q3 | Feature-based | Characteristics of Models

Which of the following is a key characteristic of a good data model?

(A) It should represent data structure, relationships, and constraints clearly

(B) It should avoid representing relationships between data

(C) It should depend on a specific hardware platform

(D) It should be ambiguous to allow flexibility

Correct Answer: A

Explanation: A good data model clearly and unambiguously represents the structure of data, the relationships among data items, and the rules (constraints) governing them.

Q4 | Concept-based | Importance of Modeling

Why is data modeling considered important in database system design?

A. It helps identify data requirements early and reduces design errors

B. It automatically generates hardware specifications

C. It eliminates the need for testing

D. It removes the need for a database administrator

Correct Answer: A

Explanation: Modeling early in the design process helps catch inconsistencies and missing requirements before costly implementation, reducing errors.

Q5 | Application-oriented | Applications of Models

Data models are primarily applied in database systems to:

A. Represent real-world entities and their relationships for database design

B. Replace application software

C. Manage electricity consumption of servers

D. Design network cables

Correct Answer: A

Explanation: Data models translate real-world objects and their interconnections into a structured form usable for designing and implementing databases.

Q6 | Statement-based | Definition of Model

Statement: A model provides an abstract representation of real-world data.
Is this statement correct?

A. False, because models must always represent physical hardware

B. True, but only for hierarchical models

C. False, because models cannot represent real-world data

D. True, because models abstract complex real-world details into a simplified structure

Correct Answer: D

Explanation: All data models act as abstractions, simplifying complex real-world information into structured, understandable forms.

Q7 | Assertion-Reason | Characteristics of Models

Assertion (A): A data model must clearly define constraints on data.
Reason (R): Constraints ensure data integrity and consistency within the database.

A. A is true, but R is false

B. Both A and R are true, but R does not explain A

C. Both A and R are true, and R correctly explains A

D. A is false, but R is true

Correct Answer: C

Explanation: Constraints are a mandatory part of any data model because they enforce integrity and consistency, which directly explains why models must define them.

Q8 | Tricky | Purpose of Data Modeling

Which of the following is NOT a purpose of data modeling?

A. Reducing redundancy through structured design

B. Providing a communication tool between designers and users

C. Understanding data requirements

D. Physically manufacturing storage hardware

Correct Answer: D

Explanation: Data modeling is a conceptual and logical activity; it has nothing to do with manufacturing physical hardware.

Q9 | Scenario-based | Applications of Models

A college wants to design a database to manage students, courses, and faculty. The first step recommended is to:

A Directly write SQL queries

B Purchase new servers

C Create a data model representing entities and their relationships

D Install a network model exam directly

Correct Answer: C

Explanation: Before implementation, designers should first create a data model to represent the entities (students, courses, faculty) and their relationships.

Q10 | Basic | Characteristics of Models

Which characteristic allows a data model to be implemented across different database systems?

A Fixed screen resolution

B Dependency on a single vendor's software

C Hardware dependency

D Abstraction and independence from physical implementation

Correct Answer: D

Explanation: A good data model is abstract and implementation-independent, allowing it to be realized on different DBMS platforms.

Q11 | Previous-Year Pattern | Importance of Modeling

In competitive exams, data modeling is often described as the _____ step of database design.

A Foundational/First

B Optional

C Unnecessary

D Final

Correct Answer: A

Explanation: Data modeling is the foundational first step of database design, guiding all later logical and physical design decisions.

Q12 | Match-the-Following | Definition of Model

Match the term with its correct description:
1. Model 2. Data Modeling 3. Constraint 4. Entity
A. Rule limiting data values B. Real-world object represented in data C. Abstract representation of data D. Process of creating a data structure blueprint

(A) 1-A, 2-B, 3-C, 4-D

(B) 1-D, 2-A, 3-B, 4-C

(C) 1-B, 2-C, 3-D, 4-A

(D) 1-C, 2-D, 3-A, 4-B

Correct Answer: D

Explanation: Model = abstract representation (C); Data Modeling = process of creating the blueprint (D); Constraint = rule limiting values (A); Entity = real-world object (B).

Q13 | Difficult | Applications of Models

Which statement best justifies the use of data models before database implementation?

(A) They are only useful for very large databases

(B) They allow early detection of design flaws and align stakeholder understanding

(C) They guarantee zero errors in the final database

(D) They eliminate the need for a DBMS

Correct Answer: B

Explanation: Data models act as a communication and validation tool, allowing designers and stakeholders to detect flaws early, regardless of database size.

Q14 | Concept-based | Characteristics of Models

A data model that can represent complex relationships while remaining easy to understand is said to have:

(A) No constraints

(B) Redundant structure

(C) Balanced simplicity and expressiveness

(D) Poor abstraction

Correct Answer: C

Explanation: An effective model balances simplicity (ease of understanding) with expressiveness (ability to represent complex relationships).

Q15 | Basic | Purpose of Data Modeling

Data modeling helps reduce which of the following in a database?

(A) Data redundancy and inconsistency

(B) Number of users

(C) Number of tables always

(D) Hardware cost only

Correct Answer: A

Explanation: Proper data modeling identifies relationships and structure clearly, which helps minimize redundancy and inconsistency in the database.

Chapter 2: Hierarchical Model

12 Questions

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Q16 | Definition-based | Structure

The Hierarchical data model organizes data in the form of a:

(A) Random network

(B) Graph with no root

(C) Flat table

(D) Tree structure with parent-child relationships

Correct Answer: D

Explanation: The Hierarchical Model arranges records in a tree-like structure where each child record has exactly one parent.

Q17 | Concept-based | Structure

In the Hierarchical Model, the relationship between parent and child records is:

(A) One-to-many

(B) No relationship

(C) Many-to-one only

(D) Many-to-many

Correct Answer: A

Explanation: In the hierarchical model, one parent record can have multiple child records, forming a one-to-many relationship.

Q18 | Feature-based | Characteristics

Which of the following is a key feature of the Hierarchical Model?

(A) Each child record has only one parent record

(B) It uses SQL as its only query mechanism

(C) Each child can have multiple parents

(D) Data is stored without any structure

Correct Answer: A

Explanation: A defining feature of the hierarchical model is that every child node has exactly one parent node, unlike the network model.

Q19 | Concept-based | Advantages

An advantage of the Hierarchical Model is:

A. It eliminates the need for a root node

B. It requires no predefined structure

C. It provides fast access when relationships are naturally tree-structured

D. It handles many-to-many relationships efficiently

Correct Answer: C

Explanation: Because it mirrors naturally hierarchical relationships (like organizational charts), data access along the tree is efficient.

Q20 | Concept-based | Limitations

A major limitation of the Hierarchical Model is:

A. It cannot represent one-to-many relationships

B. Difficulty in representing many-to-many relationships

C. It cannot be used in any database

D. It has no root node

Correct Answer: B

Explanation: Since each child can have only one parent, representing many-to-many relationships in the hierarchical model is complex and often requires data duplication.

Q21 | Application-oriented | Applications

The Hierarchical Model is most suitable for applications such as:

A. Social network friend connections

B. E-commerce recommendation graphs

C. Distributed peer-to-peer networks

D. Organizational charts and file directory systems

Correct Answer: D

Explanation: Applications with a natural tree structure, like organizational hierarchies or file systems, are well suited to the hierarchical model.

Q22 | Statement-based | Structure

Statement: In the Hierarchical Model, the topmost node is called the root node.
Is this correct?

A. True
B. False, root exists only in relational model
C. True, only in network model
D. False

Correct Answer: A

Explanation: In tree-structured hierarchical models, the topmost node with no parent is indeed called the root node.

Q23 | Tricky | Limitations

Which problem is most associated with the Hierarchical Model when a child logically belongs to two parents?

A. The model automatically creates a many-to-many link
B. The DBMS crashes
C. No problem occurs; it is handled natively
D. Data duplication or awkward workarounds are required

Correct Answer: D

Explanation: Since only one parent per child is allowed, representing a child under two parents forces data duplication or artificial restructuring.

Q24 | Basic | Examples

Which of the following is a classic early example of a Hierarchical database system?

(A) MongoDB

(B) PostgreSQL

(C) MySQL

(D) IBM's IMS (Information Management System)

Correct Answer: D

Explanation: IBM's IMS, developed in the 1960s, is a well-known example of a database system based on the hierarchical model.

Q25 | Assertion-Reason | Structure

Assertion (A): The Hierarchical Model is represented as an inverted tree.
Reason (R): The root node appears at the top with child nodes branching downward.

(A) A is true, R is false

(B) Both A and R are true, and R correctly explains A

(C) Both A and R are true, but R does not explain A

(D) A is false, R is true

Correct Answer: B

Explanation: The tree is called 'inverted' because the root sits at the top with branches extending downward, which is exactly what R explains.

Q26 | Previous-Year Pattern | Advantages

In the Hierarchical Model, data retrieval along the tree path is generally:

(A) Fast, since access follows a predictable parent-child path

(B) Dependent only on network bandwidth

(C) Slow due to random access

(D) Impossible without SQL

Correct Answer: A

Explanation: Because relationships are predefined as tree paths, navigating from parent to child is fast and predictable.

Q27 | Scenario-based | Limitations

A library wants to store books where each book can belong to multiple categories. Using a strict Hierarchical Model would cause:

(A) Faster retrieval than any other model

(B) Automatic support for multiple categories

(C) Difficulty since each record can have only one parent category

(D) No issues at all

Correct Answer: C

Explanation: Since a book (child) needing multiple categories (parents) violates the one-parent rule, the hierarchical model struggles with this scenario.

Chapter 3: Network Model

12 Questions

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Q28 | Definition-based | Structure

The Network Model represents data using:

(A) A graph structure allowing a child to have multiple parents

(B) No relationships between records

(C) A strict tree with one parent per child

(D) Only two-dimensional tables

Correct Answer: A

Explanation: Unlike the hierarchical model, the network model uses a graph structure where a child (member) record can be linked to multiple parent (owner) records.

Q29 | Feature-based | Characteristics

In the Network Model, relationships between records are represented using:

(A) Class hierarchies

(B) Owner-member sets (pointers/links)

(C) Rows and columns only

(D) XML tags

Correct Answer: B

Explanation: The network model uses 'owner-member' set relationships, implemented through pointers, to connect related records.

Q30 | Concept-based | Structure

Compared to the Hierarchical Model, the Network Model allows:

(A) Only one-to-one relationships

(B) Many-to-many relationships between records

(C) No relationships at all

(D) Only self-referencing relationships

Correct Answer: B

Explanation: A key advancement of the network model is its ability to represent many-to-many relationships directly, which the hierarchical model cannot easily do.

Q31 | Concept-based | Advantages

An advantage of the Network Model over the Hierarchical Model is:

A. No need for pointers

B. Simpler structure with fewer relationships

C. Elimination of the owner-member concept

D. Greater flexibility in representing complex relationships

Correct Answer: D

Explanation: Because it supports many-to-many relationships via multiple owner-member sets, the network model is more flexible than the hierarchical model.

Q32 | Concept-based | Limitations

A major limitation of the Network Model is:

A. It cannot represent any relationships

B. It cannot be implemented in any DBMS

C. It only supports one record type

D. Its complex pointer structure makes it difficult to design and maintain

Correct Answer: D

Explanation: The heavy use of pointers and set structures makes the network model complex to design, navigate, and maintain compared to later models.

Q33 | Application-oriented | Applications

The Network Model was historically well suited for applications requiring:

A. Simple flat file storage

B. Complex interlinked relationships such as engineering design systems

C. Purely tabular reporting

D. Single-user desktop notes

Correct Answer: B

Explanation: Its ability to model complex, interlinked relationships made the network model suitable for applications like CAD/engineering systems.

Q34 | Basic | Examples

Which of these is a well-known example associated with the Network Model?

A) Oracle RDBMS

B) SQLite

C) IDMS (Integrated Database Management System)

D) MongoDB

Correct Answer: C

Explanation: IDMS is a classic example of a database management system built on the network model.

Q35 | Statement-based | Structure

Statement: In the Network Model, a member record can be associated with more than one owner record.
Is this correct?

A) False

B) True, but only in hierarchical models

C) True

D) False, only one owner allowed always

Correct Answer: C

Explanation: This is precisely what distinguishes the network model from the hierarchical model — a member can belong to multiple owners.

Q36 | Tricky | Limitations

Navigating the Network Model typically requires:

A) Procedural navigation using pointers, which is complex for end users

B) Automatic natural language queries

C) No relationships at all

D) No programming knowledge, only simple menus

Correct Answer: A

Explanation: Since data is linked via pointers in sets, retrieving data requires procedural, pointer-based navigation, which is complex compared to declarative query languages.

Q37 | Assertion-Reason | Advantages

Assertion (A): The Network Model is more flexible than the Hierarchical Model.
Reason (R): It allows a record to participate in multiple relationships as a member of several sets.

(A) Both A and R are true, but R does not explain A

(B) Both A and R are true, and R correctly explains A

(C) A is true, R is false

(D) A is false, R is true

Correct Answer: B

Explanation: The flexibility of the network model directly stems from allowing records to belong to multiple sets, which R correctly explains.

Q38 | Previous-Year Pattern | Structure

In Network Model terminology, a set relationship consists of:

(A) Only member records

(B) Only tables and columns

(C) One owner record type and one or more member record types

(D) Only foreign keys

Correct Answer: C

Explanation: A 'set' in the network model defines a relationship between one owner record type and one or more associated member record types.

Q39 | Scenario-based | Limitations

A company's design system needs a part to be used in multiple assemblies (many-to-many). Which model is naturally more suited than the Hierarchical Model?

(A) Flat file model only

(B) Network Model

(C) Hierarchical Model handles it equally well

(D) No model can handle this

Correct Answer: B

Explanation: Since parts belonging to multiple assemblies is a many-to-many relationship, the Network Model, which supports multiple owners per member, is naturally more suited.

Chapter 4: Relational Model

12 Questions

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Q40 | Definition-based | Structure

The Relational Model organizes data into:

(A) Objects with inheritance

(B) Tables (relations) consisting of rows and columns

(C) Pointer-based networks

(D) Trees with parent-child nodes

Correct Answer: B

Explanation: The Relational Model, proposed by E.F. Codd, stores data in tables called relations, made up of rows (tuples) and columns (attributes).

Q41 | Feature-based | Characteristics

In the Relational Model, each row of a table is also known as a:

(A) Schema only

(B) Tuple

(C) Attribute

(D) Domain

Correct Answer: B

Explanation: A row in a relational table represents a single record and is formally called a tuple.

Q42 | Feature-based | Characteristics

In the Relational Model, each column of a table represents a(n):

(A) Relationship set

(B) Attribute (field)

(C) Tuple

(D) Owner record

Correct Answer: B

Explanation: A column in a relational table corresponds to an attribute, describing a specific property of the entity.

Q43 | Concept-based | Advantages

A major advantage of the Relational Model is:

(A) It is limited to hierarchical relationships only

(B) Data independence and ease of use through declarative query languages like SQL

(C) It cannot represent relationships between tables

(D) It requires pointer navigation for every query

Correct Answer: B

Explanation: The relational model separates logical structure from physical storage and allows powerful declarative querying via SQL, unlike pointer-based models.

Q44 | Concept-based | Limitations

A limitation sometimes associated with the basic Relational Model is:

(A) Inability to enforce any constraints

(B) Inability to represent tabular data

(C) Lack of any standard query language

(D) Reduced efficiency for highly complex, nested, or object-oriented data structures

Correct Answer: D

Explanation: While powerful for tabular data, the basic relational model can be less efficient for complex nested structures compared to object-oriented approaches.

Q45 | Application-oriented | Applications

The Relational Model is most widely used today in applications such as:

(A) Only scientific simulations

(B) Banking, inventory, and enterprise information systems

(C) Only real-time embedded control systems

(D) Only image processing

Correct Answer: B

Explanation: Due to its structured, reliable, and query-friendly nature, the relational model dominates enterprise applications like banking and inventory systems.

Q46 | Basic | Examples

Which of the following is an example of a Relational Database Management System (RDBMS)?

(A) Neo4j

(B) MongoDB

(C) IMS

(D) MySQL

Correct Answer: D

Explanation: MySQL is a widely used RDBMS that implements the relational model with tables, rows, and columns.

Q47 | Statement-based | Structure

Statement: In the Relational Model, the order of rows and columns in a table has no significance.
Is this correct?

(A) False, order always matters

(B) False

(C) True, only for columns

(D) True

Correct Answer: D

Explanation: A fundamental property of relations in the relational model is that the order of rows and columns is logically insignificant.

Q48 | Tricky | Characteristics

Which of the following best distinguishes a relational table from a spreadsheet?

(A) A spreadsheet always has more rows

(B) A relational table cannot contain numbers

(C) A relational table enforces a consistent data type per column and integrity rules

(D) There is no real difference

Correct Answer: C

Explanation: Unlike a general spreadsheet, a relational table enforces a defined schema, consistent column data types, and integrity constraints.

Q49 | Assertion-Reason | Advantages

Assertion (A): The Relational Model is the most widely adopted database model. Reason (R): It offers a simple tabular structure combined with a powerful standardized query language (SQL).

(A) A is false, R is true

(B) Both A and R are true, and R correctly explains A

(C) Both A and R are true, but R does not explain A

(D) A is true, R is false

Correct Answer: B

Explanation: Its widespread adoption is directly due to the combination of simplicity and the power of SQL, which R correctly explains.

Q50 | Previous-Year Pattern | Structure

Which term describes a table in the Relational Model formally?

(A) Set

(B) Owner

(C) Segment

(D) Relation

Correct Answer: D

Explanation: In relational database terminology, a table is formally referred to as a 'relation'.

Q51 | Scenario-based | Applications

A bank needs to store customer accounts with structured, consistent records and support complex queries. The most suitable model is:

A None of the above

B Network Model

C Relational Model

D Hierarchical Model

Correct Answer: C

Explanation: The Relational Model's structured tables, integrity constraints, and SQL query support make it ideal for structured banking data.

Chapter 5: Object-Oriented Model

10 Questions

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Q52 | Definition-based | Structure

The Object-Oriented Model represents data as:

(A) Objects that encapsulate both data (attributes) and behavior (methods)

(B) Purely hierarchical trees with no attributes

(C) Simple pointer-linked records only

(D) Rows and columns only

Correct Answer: A

Explanation: The Object-Oriented Model stores data as objects, combining attributes (data) and methods (behavior) within a single unit, similar to object-oriented programming.

Q53 | Feature-based | Characteristics

Which concept from object-oriented programming is also used in the Object-Oriented Data Model?

(A) Owner-member sets

(B) Parent-child tree paths only

(C) Row-column mapping only

(D) Inheritance and encapsulation

Correct Answer: D

Explanation: The Object-Oriented Model borrows core OOP concepts such as inheritance (class hierarchies) and encapsulation (bundling data with methods).

Q54 | Concept-based | Advantages

An advantage of the Object-Oriented Model is:

(A) It cannot represent complex data types

(B) It does not support inheritance

(C) It is restricted to simple tabular data only

(D) It efficiently represents complex data such as multimedia and nested structures

Correct Answer: D

Explanation: Because objects can hold complex, nested, and varied data types, this model handles multimedia and complex structures more naturally than tables.

Q55 | Concept-based | Limitations

A limitation of the Object-Oriented Model is:

A. Complete absence of a query mechanism

B. Lack of support for complex data

C. Inability to store attributes

D. Relatively higher complexity and less standardization compared to the relational model

Correct Answer: D

Explanation: The Object-Oriented Model, while powerful, is generally more complex to design and lacks the widespread standardization SQL provides for relational systems.

Q56 | Application-oriented | Applications

The Object-Oriented Model is particularly suitable for applications involving:

A. Simple payroll tables

B. Simple contact lists

C. CAD/CAM systems, multimedia, and complex engineering designs

D. Basic attendance registers

Correct Answer: C

Explanation: Complex applications with rich, nested data (like CAD/CAM or multimedia) benefit from the object-oriented model's ability to represent objects directly.

Q57 | Statement-based | Characteristics

Statement: In the Object-Oriented Model, objects with similar attributes and methods are grouped into classes.
Is this correct?

A True

B False

C True, but only for relational tables

D False, grouping is not possible

Correct Answer: A

Explanation: Just as in object-oriented programming, objects sharing structure and behavior are grouped into classes in the object-oriented data model.

Q58 | Tricky | Structure

Which of the following differentiates the Object-Oriented Model from the Relational Model?

A It combines data and behavior (methods) within objects, unlike relational tables

B It cannot store any data

C It uses only rows and columns

D It has no concept of attributes

Correct Answer: A

Explanation: The defining difference is encapsulation: objects bundle data and the operations (methods) that act on that data, unlike passive relational tables.

Q59 | Basic | Examples

Which of the following is associated with Object-Oriented Database Systems?

A IMS

B IDMS

C MySQL

D ObjectStore / db4o

Correct Answer: D

Explanation: Systems like ObjectStore and db4o are examples of object-oriented database management systems implementing this model.

Q60 | Assertion-Reason | Advantages

Assertion (A): The Object-Oriented Model is well suited for storing complex, structured multimedia data.
Reason (R): Objects can encapsulate varied data types and behaviors within a single unit.

(A) A is false, R is true

(B) Both A and R are true, and R correctly explains A

(C) A is true, R is false

(D) Both A and R are true, but R does not explain A

Correct Answer: B

Explanation: The suitability for complex data directly results from the encapsulation of varied data types and behavior, which R explains correctly.

Q61 | Scenario-based | Applications

A software company is designing a system to store 3D engineering models with complex nested components. The most suitable model is:

(A) Network Model

(B) Object-Oriented Model

(C) None of the above

(D) Hierarchical Model

Correct Answer: B

Explanation: Since 3D models involve complex, nested, and behavior-rich data, the Object-Oriented Model is best suited for such representation.

Chapter 6: Comparison of Data Models

10 Questions

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Q62 | Comparison-based | Comparison

Which model uses a strict one-parent-per-child rule, unlike the Network Model?

(A) None of the above

(B) Object-Oriented Model

(C) Hierarchical Model

(D) Relational Model

Correct Answer: C

Explanation: The Hierarchical Model restricts each child to a single parent, while the Network Model allows multiple parents (owners) per member.

Q63 | Comparison-based | Comparison

Which two models both rely heavily on pointer-based navigation rather than declarative queries?

(A) Hierarchical and Network

(B) Object-Oriented and Relational

(C) Relational and Object-Oriented

(D) Relational and Hierarchical

Correct Answer: A

Explanation: Both the Hierarchical and Network Models rely on pointer/link-based navigation, whereas the Relational Model uses declarative SQL queries.

Q64 | Match-the-Following | Comparison

Match the model with its core structural concept:
1. Hierarchical 2. Network 3. Relational 4. Object-Oriented
A. Owner-member sets B. Tables of rows and columns C. Parent-child tree D. Objects with methods

A 1-C, 2-A, 3-B, 4-D

B 1-A, 2-B, 3-C, 4-D

C 1-B, 2-D, 3-A, 4-C

D 1-D, 2-C, 3-A, 4-B

Correct Answer: A

Explanation: Hierarchical = tree (C); Network = owner-member sets (A); Relational = tables (B); Object-Oriented = objects with methods (D).

Q65 | Difficult | Comparison

Which model was primarily introduced to overcome the many-to-many relationship limitation of the Hierarchical Model?

A None of the above

B Relational Model

C Object-Oriented Model

D Network Model

Correct Answer: D

Explanation: The Network Model was developed specifically to allow many-to-many relationships, which the Hierarchical Model could not handle well.

Q66 | Concept-based | Comparison

Which model introduced data independence as a major design goal, separating logical and physical structure more formally than earlier models?

A Relational Model

B Network Model

C Hierarchical Model

D None of the above

Correct Answer: A

Explanation: E.F. Codd's Relational Model emphasized data independence, decoupling logical data representation from physical storage details.

Q67 | Comparison-based | Comparison

Which model is generally considered the most user-friendly for writing complex, ad-hoc queries?

(A) Network Model

(B) Relational Model (via SQL)

(C) Hierarchical Model

(D) None support ad-hoc queries

Correct Answer: B

Explanation: The Relational Model's SQL provides an easy, declarative way to write ad-hoc queries, unlike the procedural navigation in older models.

Q68 | Previous-Year Pattern | Comparison

Historically, which sequence correctly represents the evolution of data models?

(A) Network → Relational → Hierarchical → Object-Oriented

(B) Hierarchical → Network → Relational → Object-Oriented

(C) Object-Oriented → Relational → Network → Hierarchical

(D) Relational → Hierarchical → Network → Object-Oriented

Correct Answer: B

Explanation: Data models evolved historically in the order Hierarchical, then Network, followed by the Relational Model, and later the Object-Oriented Model.

Q69 | Selection of Suitable Model | Comparison

For storing simple, structured, tabular business data such as employee payroll, which model is typically the best choice today?

(A) Hierarchical Model

(B) Object-Oriented Model

(C) Network Model

(D) Relational Model

Correct Answer: D

Explanation: Simple structured tabular business data is most efficiently and reliably managed using the Relational Model with SQL support.

Q70 | Tricky | Comparison

Which statement correctly compares the Hierarchical and Relational Models?

(A) The Hierarchical Model uses tree structures, while the Relational Model uses independent tables linked logically

(B) Both use tables of rows and columns identically

(C) The Relational Model cannot represent one-to-many relationships

(D) The Hierarchical Model uses SQL as its native query language

Correct Answer: A

Explanation: The Hierarchical Model is tree-based with rigid parent-child links, while the Relational Model uses independent tables connected via key-based logical relationships.

Q71 | Assertion-Reason | Comparison

Assertion (A): The Object-Oriented Model is considered more powerful than the Relational Model for complex data.
Reason (R): It can encapsulate both data and behavior in a single structure.

(A) A is true, R is false

(B) Both A and R are true, but R does not explain A

(C) A is false, R is true

(D) Both A and R are true, and R correctly explains A

Correct Answer: D

Explanation: The added power for complex data comes precisely from encapsulating data and behavior together, which R correctly explains.

Chapter 7: Introduction to ERD

8 Questions

OAVS TGT Computer Science — Modeling Techniques & ERD Syllabus

Q72 | Definition-based | Definition of ERD

An Entity Relationship Diagram (ERD) is best defined as:

(A) A graphical representation of entities and their relationships in a database

(B) A physical wiring diagram of servers

(C) A programming flowchart

(D) A network topology diagram

Correct Answer: A

Explanation: An ERD is a diagrammatic tool used to visually represent entities, their attributes, and the relationships between them in a database design.

Q73 | Concept-based | Purpose of ERD

The main purpose of creating an ERD during database design is to:

(A) Replace the DBMS software

(B) Store actual data records

(C) Encrypt the database

(D) Visually plan and communicate the data structure before implementation

Correct Answer: D

Explanation: An ERD serves as a planning and communication tool, helping designers visualize entities and relationships before building the actual database.

Q74 | Concept-based | Importance of ERD

ERDs are considered important in database design mainly because they:

(A) Are required only for small databases

(B) Increase database size

(C) Eliminate the need for normalization entirely

(D) Help identify entities, attributes, and relationships clearly, reducing design errors

Correct Answer: D

Explanation: By clearly mapping entities and relationships early, ERDs help reduce ambiguity and design errors before physical implementation.

Q75 | Statement-based | Definition of ERD

Statement: An ERD is created during the conceptual/logical design phase of a database. Is this correct?

A. False, ERDs are used only after implementation

B. True

C. False

D. True, but only for object-oriented databases

Correct Answer: B

Explanation: ERDs are primarily conceptual and logical design tools, used before physical implementation of the database.

Q76 | Previous-Year Pattern | Purpose of ERD

Who is generally credited with introducing the Entity-Relationship model?

A. Charles Bachman

B. Edgar Allan

C. Peter Chen

D. E.F. Codd

Correct Answer: C

Explanation: Peter Chen introduced the Entity-Relationship model in 1976 as a way to conceptually design databases.

Q77 | Tricky | Importance of ERD

Which of the following is NOT a benefit of using an ERD?

A. Easier communication between designers and stakeholders

B. Early identification of missing entities or attributes

C. Automatic physical storage optimization on disk

D. Clear visualization of data relationships

Correct Answer: C

Explanation: ERDs are conceptual tools; they do not automatically optimize physical disk storage, which is a separate physical design concern.

Q78 | Basic | Definition of ERD

An ERD primarily uses which of the following to represent entities?

A. Rectangles (boxes)

B. Arrows only

C. Dotted lines only

D. Circles only

Correct Answer: A

Explanation: In standard ER notation, entities are typically represented using rectangles (boxes).

Q79 | Scenario-based | Purpose of ERD

A hospital wants to design a database for patients, doctors, and appointments. The best first step is to:

A. Write SQL CREATE TABLE statements directly

B. Skip planning and directly enter patient data

C. Draw an ERD to represent patients, doctors, appointments, and their relationships

D. Install a network model DBMS

Correct Answer: C

Explanation: Before creating tables, designers should first draw an ERD to visually plan the entities (patients, doctors, appointments) and their relationships.

Chapter 8: ERD Components: Entity & Attributes

12 Questions

OAVS TGT Computer Science — Modeling Techniques & ERD Syllabus

Q80 | Definition-based | Entity

An Entity in an ERD represents:

(A) A relationship between two tables

(B) A specific value stored in a table

(C) A query written in SQL

(D) A real-world object or concept about which data is stored

Correct Answer: D

Explanation: An entity is a real-world object or concept (like 'Student' or 'Book') about which data is collected and stored in the database.

Q81 | Definition-based | Entity Set

An Entity Set refers to:

(A) A relationship connecting two entities

(B) A single instance of an entity

(C) An attribute of an entity

(D) A collection of similar entities of the same type

Correct Answer: D

Explanation: An entity set is a collection of entities of the same type sharing the same set of attributes, e.g., all 'Student' entities form the Student entity set.

Q82 | Definition-based | Attribute

An Attribute in an ERD describes:

(A) A property or characteristic of an entity

(B) The degree of a relationship

(C) A relationship between two entity sets

(D) A diagram symbol for entities

Correct Answer: A

Explanation: An attribute represents a specific property or characteristic of an entity, such as 'Name' or 'Age' for a Student entity.

Q83 | Concept-based | Types of Attributes

An attribute that cannot be divided further into smaller sub-parts is called a:

A. Composite attribute

B. Multivalued attribute

C. Simple (atomic) attribute

D. Derived attribute

Correct Answer: C

Explanation: A simple or atomic attribute is one that cannot logically be broken down further, such as 'Age'.

Q84 | Concept-based | Types of Attributes

An attribute that can be divided into smaller sub-parts, such as 'Name' into 'First Name' and 'Last Name', is called a:

A. Single-valued attribute

B. Key attribute

C. Simple attribute

D. Composite attribute

Correct Answer: D

Explanation: A composite attribute is made up of multiple smaller attributes, e.g., 'Address' can be split into 'Street', 'City', and 'Pincode'.

Q85 | Concept-based | Types of Attributes

An attribute that can hold multiple values for a single entity, such as 'Phone Numbers', is called a:

A. Simple attribute

B. Composite attribute

C. Derived attribute

D. Multivalued attribute

Correct Answer: D

Explanation: A multivalued attribute can take more than one value for a single entity, such as a person having multiple phone numbers.

Q86 | Concept-based | Types of Attributes

An attribute whose value can be calculated from other attributes, such as 'Age' from 'Date of Birth', is called a:

A) Composite attribute

B) Key attribute

C) Derived attribute

D) Multivalued attribute

Correct Answer: C

Explanation: A derived attribute's value is not stored directly but computed from other attributes, like calculating Age from Date of Birth.

Q87 | Statement-based | Entity

Statement: A specific instance, like 'Ravi' as a student, is called an entity, while 'Student' as a category is the entity set.
Is this correct?

A) False

B) True

C) True, but reversed terminology

D) False, both terms mean the same thing

Correct Answer: B

Explanation: This correctly distinguishes an individual entity instance ('Ravi') from the entity set (the collection 'Student').

Q88 | Match-the-Following | Attribute

Match the attribute type with its example:
1. Simple 2. Composite 3. Multivalued 4. Derived
A. Age (calculated from DOB) B. Phone numbers (more than one) C. Roll Number D. Address (Street, City, Pin)

(A) 1-A, 2-B, 3-C, 4-D

(B) 1-B, 2-A, 3-D, 4-C

(C) 1-D, 2-C, 3-A, 4-B

(D) 1-C, 2-D, 3-B, 4-A

Correct Answer: D

Explanation: Simple = Roll Number (C); Composite = Address (D); Multivalued = Phone numbers (B); Derived = Age from DOB (A).

Q89 | Tricky | Entity

Which of the following is generally considered a 'weak entity' concept relevant even at an introductory ERD level?

(A) An entity that cannot participate in any relationship

(B) An entity with no attributes at all

(C) An entity that has its own unique key and exists independently

(D) An entity that depends on another entity for its identification

Correct Answer: D

Explanation: A weak entity cannot be uniquely identified by its own attributes alone and depends on a related (owner) entity for identification.

Q90 | Assertion-Reason | Entity Set

Assertion (A): All entities within an entity set share the same set of attributes.
Reason (R): An entity set is defined as a grouping of entities of the same type.

(A) A is true, R is false

(B) Both A and R are true, and R correctly explains A

(C) A is false, R is true

(D) Both A and R are true, but R does not explain A

Correct Answer: B

Explanation: Because an entity set groups entities of the same type, they naturally share a common attribute structure, which R explains.

Q91 | Scenario-based | Attribute

For a 'Book' entity, which attribute would most appropriately be modeled as multivalued?

(A) Publication Year

(B) ISBN

(C) Author(s), since a book may have more than one author

(D) Title

Correct Answer: C

Explanation: Since a book can have multiple authors, the Author attribute is best modeled as multivalued.

Chapter 9: Relationships, Degree & Cardinality

14 Questions

OAVS TGT Computer Science — Modeling Techniques & ERD Syllabus

Q92 | Definition-based | Relationship

A Relationship in an ERD represents:

(A) A symbol used only for entities

(B) A type of attribute

(C) An association between two or more entities

(D) A property of a single entity

Correct Answer: C

Explanation: A relationship shows how two or more entities are associated with each other, such as 'Student ENROLLS IN Course'.

Q93 | Definition-based | Degree of Relationship

The Degree of a Relationship refers to:

(A) The number of rows in a table

(B) The number of entity sets participating in the relationship

(C) The number of attributes in an entity

(D) The strength of a foreign key

Correct Answer: B

Explanation: Degree refers to how many entity sets take part in a given relationship, e.g., a binary relationship involves two entity sets.

Q94 | Concept-based | Degree of Relationship

A relationship involving exactly two entity sets, such as 'Employee WORKS IN Department', is called a:

(A) Unary relationship

(B) Ternary relationship

(C) Quaternary relationship

(D) Binary relationship

Correct Answer: D

Explanation: A binary relationship connects exactly two entity sets, which is the most common type of relationship in ERDs.

Q95 | Concept-based | Degree of Relationship

A relationship involving only one entity set, such as 'Employee SUPERVISES Employee', is called a:

A. Binary relationship

B. Composite relationship

C. Unary (recursive) relationship

D. Ternary relationship

Correct Answer: C

Explanation: A unary or recursive relationship connects an entity set to itself, such as an employee supervising another employee.

Q96 | Definition-based | Cardinality

Cardinality in an ERD refers to:

A. The name given to a relationship

B. The number of attributes an entity has

C. The numeric relationship (one or many) between entities participating in a relationship

D. The degree of normalization

Correct Answer: C

Explanation: Cardinality specifies how many instances of one entity can be associated with instances of another entity in a relationship (e.g., one-to-many).

Q97 | Concept-based | Cardinality

In a 'One Department has Many Employees' relationship, the cardinality is best described as:

A Many-to-Many

B Many-to-One only

C One-to-Many

D One-to-One

Correct Answer: C

Explanation: Since a single department can have multiple employees, but each employee belongs to one department, this is a one-to-many relationship.

Q98 | Concept-based | Cardinality

A relationship where 'Many Students can enroll in Many Courses' represents which cardinality?

A One-to-Many

B Zero cardinality

C Many-to-Many

D One-to-One

Correct Answer: C

Explanation: Since students can enroll in multiple courses and courses can have multiple students, this is a many-to-many relationship.

Q99 | Concept-based | Cardinality

A relationship where each Person has exactly one Passport, and each Passport belongs to exactly one Person, is:

A Many-to-One

B One-to-Many

C One-to-One

D Many-to-Many

Correct Answer: C

Explanation: Since each person has only one passport and each passport belongs to only one person, this is a one-to-one relationship.

Q100 | Statement-based | Relationship

Statement: A relationship in an ERD can also have its own attributes.
Is this correct?

(A) True, only for unary relationships

(B) False

(C) False, relationships never have attributes

(D) True

Correct Answer: D

Explanation: Relationships can carry their own attributes, such as 'Date of Enrollment' for a Student-Course enrollment relationship.

Q101 | Match-the-Following | Degree of Relationship

Match the relationship type with its example:
1. Unary 2. Binary 3. Ternary
A. Employee-Department-Project B. Employee WORKS IN Department C. Employee MANAGES Employee

(A) 1-C, 2-A, 3-B

(B) 1-A, 2-C, 3-B

(C) 1-C, 2-B, 3-A

(D) 1-B, 2-A, 3-C

Correct Answer: C

Explanation: Unary = Employee manages Employee (C); Binary = Employee works in Department (B); Ternary = Employee-Department-Project (A).

Q102 | Tricky | Cardinality

Which cardinality notation is typically used near an entity in Chen's ERD notation to represent 'many'?

(A) N or M

(B) 1

(C) A single dot

(D) 0

Correct Answer: A

Explanation: In Chen's notation, 'N' or 'M' near an entity typically denotes the 'many' side of a relationship's cardinality.

Q103 | Assertion-Reason | Relationship

Assertion (A): The relationship 'Teaches' between Teacher and Subject is a binary relationship.
Reason (R): It connects exactly two entity sets, Teacher and Subject.

(A) Both A and R are true, but R does not explain A

(B) Both A and R are true, and R correctly explains A

(C) A is false, R is true

(D) A is true, R is false

Correct Answer: B

Explanation: The relationship is binary precisely because it connects two entity sets, Teacher and Subject, which R correctly explains.

Q104 | Scenario-based | Cardinality

In a library system, one book can be borrowed by many different members over time, but at any moment, one member borrows a specific copy. The overall borrowing relationship between Book and Member is generally modeled as:

(A) Zero relationships

(B) One-to-One

(C) Ternary only

(D) One-to-Many/Many-to-Many depending on design

Correct Answer: D

Explanation: Depending on whether copies are tracked individually, the Book-Member borrowing relationship is typically modeled as one-to-many or many-to-many.

Q105 | Previous-Year Pattern | Degree of Relationship

Which of the following best describes a ternary relationship?

A. A relationship connecting exactly three entity sets

B. A relationship connecting an entity to itself

C. A relationship with no entity sets

D. A relationship connecting exactly two entity sets

Correct Answer: A

Explanation: A ternary relationship involves exactly three entity sets participating together, such as Supplier-Part-Project.

Chapter 10: ERD Symbols & Diagram Representation

12 Questions

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Q106 | Definition-based | ERD Symbols

In standard ER notation, a diamond shape is used to represent a:

(A) Weak entity

(B) Entity

(C) Attribute

(D) Relationship

Correct Answer: D

Explanation: In Chen's ER notation, a diamond shape is the standard symbol used to represent a relationship between entities.

Q107 | Definition-based | ERD Symbols

In standard ER notation, an oval (ellipse) shape is used to represent a(n):

(A) Entity set

(B) Attribute

(C) Entity

(D) Relationship

Correct Answer: B

Explanation: An oval or ellipse shape is used in ER diagrams to represent an attribute of an entity.

Q108 | Concept-based | ERD Symbols

A double rectangle in an ER diagram typically represents a:

(A) Derived attribute

(B) Simple entity

(C) Weak entity

(D) Composite entity

Correct Answer: C

Explanation: A double-lined rectangle is the standard symbol used to represent a weak entity, which depends on another entity for identification.

Q109 | Concept-based | ERD Symbols

A dashed oval in an ER diagram is used to represent a:

A) Multivalued attribute

B) Derived attribute

C) Key attribute

D) Simple attribute

Correct Answer: B

Explanation: A dashed oval represents a derived attribute, whose value is computed from other attributes rather than stored directly.

Q110 | Concept-based | ERD Symbols

A double oval in an ER diagram is used to represent a:

A) Multivalued attribute

B) Simple attribute

C) Derived attribute

D) Key attribute

Correct Answer: A

Explanation: A double-lined oval represents a multivalued attribute, indicating that the attribute can hold more than one value.

Q111 | Concept-based | ERD Symbols

An underlined attribute name in an ER diagram indicates that the attribute is a:

A) Composite attribute

B) Multivalued attribute

C) Derived attribute

D) Key (identifying) attribute

Correct Answer: D

Explanation: Underlining an attribute's name is the standard convention to indicate it is a key attribute that uniquely identifies the entity.

Q112 | Statement-based | ER Diagram Representation

Statement: Lines in an ER diagram are used to connect entities to their relationships and attributes.
Is this correct?

A) True

B) False, lines represent entities

C) True, only for weak entities

D) False

Correct Answer: A

Explanation: Lines are the connecting elements in ER diagrams, linking entities to relationships and to their respective attributes.

Q113 | Diagram-based | ER Diagram Representation

In a simple ERD showing 'Student' -- 'Enrolls' -- 'Course', which shape correctly represents 'Enrolls'?

A) Oval

B) Rectangle

C) Diamond

D) Triangle

Correct Answer: C

Explanation: 'Enrolls' is a relationship connecting Student and Course, so it should be represented using a diamond shape.

Q114 | Interpretation of ERD | ER Diagram Representation

In an ER diagram, if 'Name' is connected to 'Employee' by a line ending in an oval, this represents:

(A) An attribute of the Employee entity

(B) A weak entity related to Employee

(C) A cardinality label

(D) A relationship between two entities

Correct Answer: A

Explanation: An oval connected directly to an entity by a line represents an attribute describing that entity, here 'Name' describing 'Employee'.

Q115 | Match-the-Following | ERD Symbols

Match the ERD symbol with what it represents:
1. Rectangle 2. Oval 3. Diamond 4. Double Rectangle
A. Relationship B. Weak Entity C. Entity D. Attribute

(A) 1-D, 2-C, 3-B, 4-A

(B) 1-B, 2-A, 3-D, 4-C

(C) 1-A, 2-B, 3-C, 4-D

(D) 1-C, 2-D, 3-A, 4-B

Correct Answer: D

Explanation: Rectangle = Entity (C); Oval = Attribute (D); Diamond = Relationship (A); Double Rectangle = Weak Entity (B).

Q116 | Tricky | ER Diagram Representation

Which of the following is TRUE regarding lines connecting a relationship to entities with cardinality labels?

(A) Cardinality labels (like 1, N) are typically placed near the connecting lines close to each entity

(B) Cardinality is never shown in a basic ERD

(C) Cardinality labels are placed only inside entity rectangles

(D) Cardinality labels replace the need for relationship diamonds

Correct Answer: A

Explanation: Cardinality indicators (such as 1 or N) are conventionally placed on or near the connecting lines, close to the respective entity.

Q117 | Previous-Year Pattern | ER Diagram Representation

The notation style using rectangles, ovals, and diamonds for ERDs was introduced by:

(A) Donald Knuth

(B) E.F. Codd

(C) Peter Chen

(D) Charles Bachman

Correct Answer: C

Explanation: Peter Chen's original ER notation uses rectangles for entities, ovals for attributes, and diamonds for relationships.

Chapter 11: Advantages & Applications of ERD

9 Questions

OAVS TGT Computer Science — Modeling Techniques & ERD Syllabus

Q118 | Concept-based | Advantages of ERD

One major advantage of using an ERD during database design is:

(A) It provides a clear, visual understanding of data structure and relationships

(B) It eliminates the need for testing the database

(C) It replaces the need for a DBMS

(D) It automatically writes SQL queries

Correct Answer: A

Explanation: ERDs offer an intuitive visual representation, making it easier for designers and stakeholders to understand data structures and relationships.

Q119 | Concept-based | Advantages of ERD

ERDs help improve communication between database designers and end users mainly because they:

(A) Hide the relationships between entities

(B) Require SQL knowledge to interpret

(C) Use simple, standardized visual symbols understandable to non-technical users

(D) Use complex programming code

Correct Answer: C

Explanation: The standardized, simple visual symbols in ERDs make it easier for both technical and non-technical stakeholders to understand the data design.

Q120 | Application-oriented | Applications of ERD

ERDs are commonly used in which phase of the database development lifecycle?

(A) Only during data entry

(B) Only during database backup

(C) Physical implementation only

(D) Conceptual and logical design phase

Correct Answer: D

Explanation: ERDs are primarily created during the conceptual and logical design phases, before physical implementation of the database.

Q121 | Statement-based | Advantages of ERD

Statement: ERDs help identify redundant or missing entities and relationships before implementation.
Is this correct?

A. True, but only after implementation

B. True

C. False, ERDs cannot identify redundancy

D. False

Correct Answer: B

Explanation: Reviewing an ERD during design helps designers spot missing entities, redundant attributes, or unclear relationships before building the database.

Q122 | Scenario-based | Applications of ERD

A software team is starting a new e-commerce project involving customers, products, and orders. Creating an ERD first would primarily help them:

A. Directly deploy the application to production

B. Avoid using a DBMS altogether

C. Skip the requirement gathering phase

D. Visually plan the entities and relationships before coding the database

Correct Answer: D

Explanation: An ERD helps the team visually plan and validate the structure of customers, products, and orders before implementation begins.

Q123 | Basic | Basic Terminology

In ERD terminology, the term 'schema' refers to:

(A) The overall logical structure/design of the database

(B) A single data value

(C) A type of relationship only

(D) A specific row of data

Correct Answer: A

Explanation: A schema represents the overall logical design or structure of the database, often derived from the ERD.

Q124 | Tricky | Advantages of ERD

Which of the following is a limitation, not an advantage, of ERDs?

(A) Very large systems can result in complex diagrams that are hard to read

(B) They improve stakeholder communication

(C) They visually represent relationships

(D) They aid early design validation

Correct Answer: A

Explanation: While useful, ERDs can become large and difficult to interpret for very complex systems with many entities and relationships.

Q125 | Previous-Year Pattern | Applications of ERD

ERDs are considered a part of which broader database design methodology?

(A) Bottom-up coding without planning

(B) Top-down (conceptual to physical) design methodology

(C) Physical design only

(D) Only used in NoSQL systems

Correct Answer: B

Explanation: ERDs support a top-down design methodology, starting from a conceptual model and gradually moving toward physical implementation.

Q126 | Assertion-Reason | Basic Terminology

Assertion (A): ERDs are considered essential tools in relational database design.
Reason (R): They help translate real-world requirements into a structured set of entities and relationships that map to tables.

(A) A is true, R is false

(B) A is false, R is true

(C) Both A and R are true, and R correctly explains A

(D) Both A and R are true, but R does not explain A

Correct Answer: C

Explanation: ERDs are essential precisely because they bridge real-world requirements and the structured relational design, which R correctly explains.